XSuM: a generative surrogate of the SBND reconstruction chain trained on the BNB Monte Carlo — technical note

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Abstract

XSuM is a per-event surrogate of the SBND simulation and reconstruction chain: given the true neutrino vertex and the post-FSI final-state particles of an interaction, it samples the record that the cafpyana analysis dataframes hold for that interaction (whether and how many Pandora slices are truth-matched to it, the slice-level block of about fifty reconstructed quantities, and later the set of particle-flow objects), in the same dataframe format as the production. Analyses then run unchanged on the surrogated dataframes. The model is trained on the 2026 gen2 BNB + cosmics central-value Monte Carlo (v10_14_02_03) processed with a selection-free cafpyana configuration written for this purpose. This note documents the data, the inputs and outputs, the baseline and flow-matching models, and their closure on held-out files. It is updated as the production and the models progress.

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1 Purpose and scope

The physics case is forward folding without re-simulation: a cross-section model that was never run through Geant4, Pandora and cafpyana (another generator, a modified FSI model, a new kinematic region) is transported to reconstruction level by the surrogate and compared with data or with the nominal MC at that level. The surrogate does not fix data/MC disagreement; it reproduces the MC detector response, including its flaws, for new inputs.

Scope decisions (2026-10-16): XSuM generates the complete reconstruction record, not analysis quantities. Selection flags, BDT scores and derived kinematics such as $|t|$ are recomputed downstream by the analysers (CC coherent pion: S. Oh; charged kaons: T. Wester) from the generated variables. Cosmic-only slices do not depend on the neutrino model and are reused from the MC. Generator-internal labels (interaction mode, CC/NC, W, Q^2, E_ν, signal flags) are never inputs; they are kept in a separate label table for holdout tests.

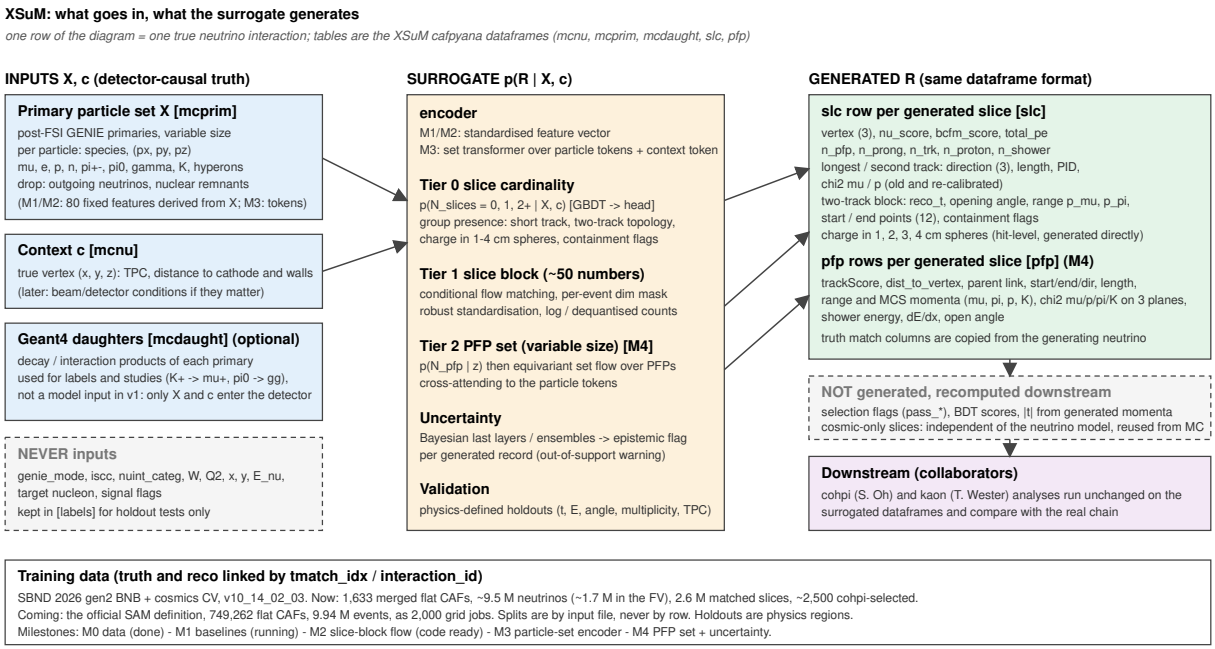


Figure 1: Inputs, surrogate, generated record and downstream use. One row of the diagram is one true neutrino interaction.

2 Data

Production. A cafpyana df-maker configuration (`analysis_village/xsum/`) writes, per flat CAF: `mcnu` (every true neutrino with the full `make_mcdf` truth, the cohpi kinematics and category), `mcprim` (every post-FSI primary), `mcdaught` (the Geant4 daughters of every primary), `slc` (every non-clear-cosmic slice with the cohpi variables and each selection stage as a flag), `pfp` (every PFP of those slices with track, shower and PID variables, kaon PID on all planes, parent links and truth match), `hdr` and POT tables. Two productions:

- **scratch1633**: the 1,633 merged flat CAFs (2.5 GB, about 3,300 triggers each) still present on the dCache scratch area referenced by the cafpyana sample list (1,363 of the 2,996 had

been purged), run in pool mode on `sbndbuild03` with 30 workers in about 3.5 h: 15.5 GB of dataframes, 8,145,755 neutrinos in the five chunks used here. This production predates the `mcdaught` table, the kaon PID branches and the removal of the 50 cm vertex-distance cut on PFPs.

- **official**: the SAM definition `aurora_SBND2026A_gen2_BNBLight_prodgenie_corsika_proton_rockbox0p1_sl` (749,262 flat CAFs, 9.94 M triggers, 7.9 TB), submitted as 2,000 grid jobs on 2026-10-16 after a 375-file pilot (21.5 s per file single-threaded, 4,500 CPU-hours in total).

Truth–reco link. A slice points to its truth-matched neutrino through `tmatch_idx`; a PFP points to the true particle it was matched to and, through `interaction_id`, to its neutrino. A neutrino can have zero, one or several matched slices; the table keeps the slice with the highest match efficiency and records the count.

Splits. Train / validation / test are 80 / 10 / 10 by input file, never by row. In the five scratch chunks: 6,514,860 / 814,028 neutrinos in train / test. Holdout tests use physics regions (momentum transfer, energy, angle, multiplicity, TPC), see Sec. 6.

3 Inputs and targets

Inputs. The post-FSI primary particles (species, (p_x, p_y, p_z); neutrinos and nuclear remnants dropped) and the true vertex. The M1 and M2 models use 80 fixed features derived from them: the vertex and its distances to the cathode and the walls; per species (mu, e, p, n, $\mathrm{pi}^{\pm}$, pi^{0}, gamma, $\mathrm{K}^{\pm}$, K^{0}, Lambda, Sigma) the multiplicity above the cohpi thresholds (27 / 30 / 20 MeV for mu / pi / p), the summed and maximum kinetic energy and the leading particle's momentum, $\cos\theta$ and p_T; and totals (charged multiplicity, visible and total kinetic energy, electromagnetic energy, longitudinal momentum). The particle-set encoder (M3) will consume the particles directly.

Targets. Tier 0: the number of matched slices and, per slice, which groups of variables exist (a second track, the two-track topology, hits within 1–4 cm of the vertex) and the two containment flags. Tier 1: the slice block (Table **??** lists the columns). Tier 2 (M4): the PFP set.

4 M1 baselines

Gradient-boosted classifiers for the discrete targets and mixture density networks (MDN, eight diagonal Gaussians, against a single Gaussian) for continuous responses, on the 80 features. Results on the test files of the five scratch chunks (reports/m1_scratch1633).

Slice efficiency. 32.1% of all true neutrinos (dirt and out-of-volume included) have a truth-matched slice. Log loss 0.1832 against a marginal 0.6270, AUC 0.976. The calibration against vertex z, muon momentum, visible energy and multiplicity is shown in Fig. 2.

cohpi selection as a yardstick. The full cohpi selection (including the sphere-charge and cathode-crossing cuts) keeps 0.035% of neutrinos. As a check of the production, the selected count per POT agrees with the technote cut flow within 10%. The classifier reaches log loss 0.00263 against 0.00313 (AUC 0.857); per true category:

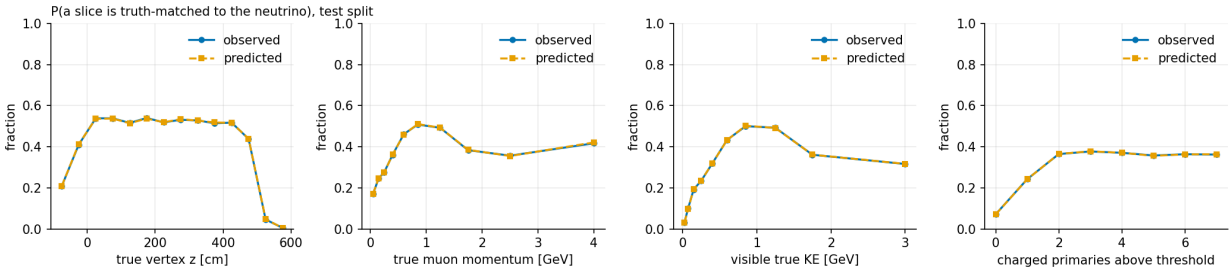


Figure 2: Observed and predicted probability that a slice is truth-matched to the neutrino, test files.

True category	test neutrinos	selected [%]	predicted [%]
out of FV	667,795	0.000	0.024
NC	41,319	0.063	0.040
signal	212	27.358	37.635
CCQE	55,425	0.020	0.021
2p2h	14,431	0.049	0.025
RES	29,189	0.483	0.500
DIS	5,657	0.672	0.815

PFP multiplicity of the matched slice: log loss 0.861 against 1.414, accuracy 66.7% against 39.0% (Fig. 3).

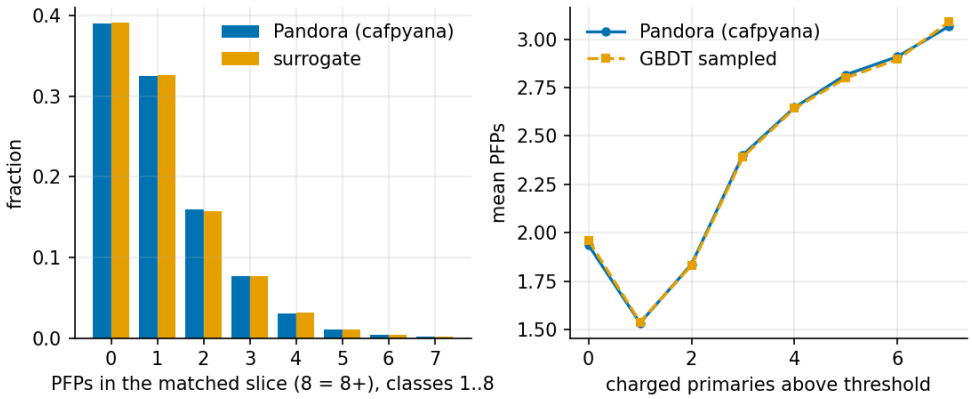


Figure 3: PFP multiplicity of the matched slice: marginal and mean versus charged primaries.

Continuous responses. Table 1 gives the test negative log-likelihood and Table 2 the marginals; Figs. 4–6 show them.

Target	training rows	Gaussian NLL	MDN ($K = 8$) NLL
vertex residual (3-d)	986,566	5.571	0.871
leading track: $\Delta \cos\theta$, log length	962,124	0.078	-2.648
two-track: log range-p ratios, $\Delta \cos\theta_\mu$	18,311	0.057	-1.710

Table 1: M1 MDN test NLL per target.

Variable	Pandora median [16%, 84%]	MDN median [16%, 84%]	W_1
dvtx_x [cm]	0.000 [-0.203, 0.210]	-0.000 [-0.207, 0.211]	2.3234
dvtx_y [cm]	0.001 [-0.381, 0.394]	0.002 [-0.359, 0.392]	2.9230
dvtx_z [cm]	-0.011 [-0.395, 0.479]	-0.010 [-0.378, 0.465]	3.9131
long_dirz - lead cos theta	-0.001 [-0.045, 0.060]	-0.002 [-0.048, 0.062]	0.0626
log(long track length [cm])	4.731 [3.669, 5.490]	4.733 [3.660, 5.495]	0.0074
log(range_p_mu / p_mu_true)	-0.013 [-0.615, 0.081]	-0.014 [-0.587, 0.157]	0.0241
log(range_p_cpi / p_pi_true)	-0.498 [-1.082, -0.136]	-0.509 [-1.060, -0.117]	0.0356
long_dirz - cos theta_mu	-0.000 [-0.030, 0.063]	0.000 [-0.027, 0.099]	0.0311

Table 2: Marginals of the MDN samples against Pandora, test files.

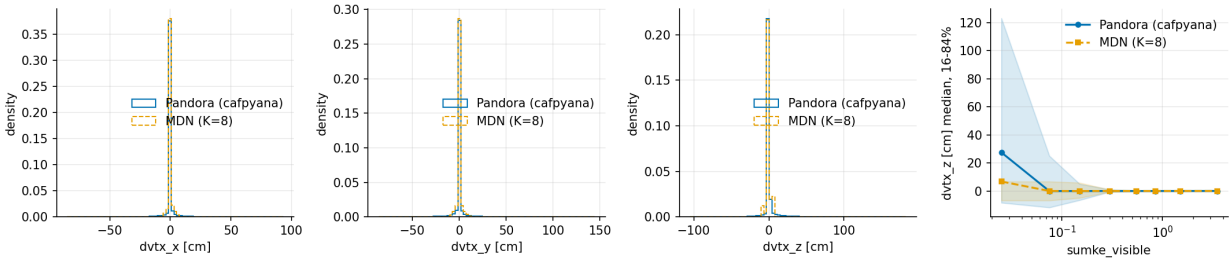


Figure 4: Vertex residual of the matched slice (FV neutrinos).

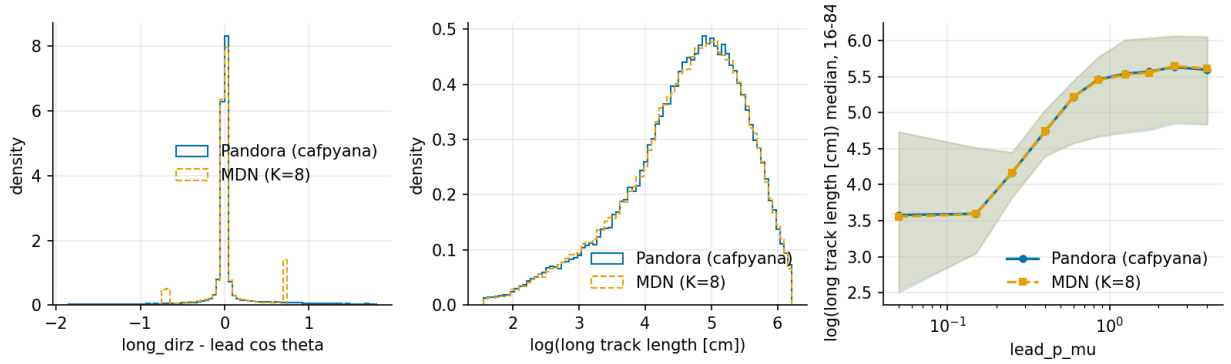


Figure 5: Longest track: direction relative to the leading charged primary and length.

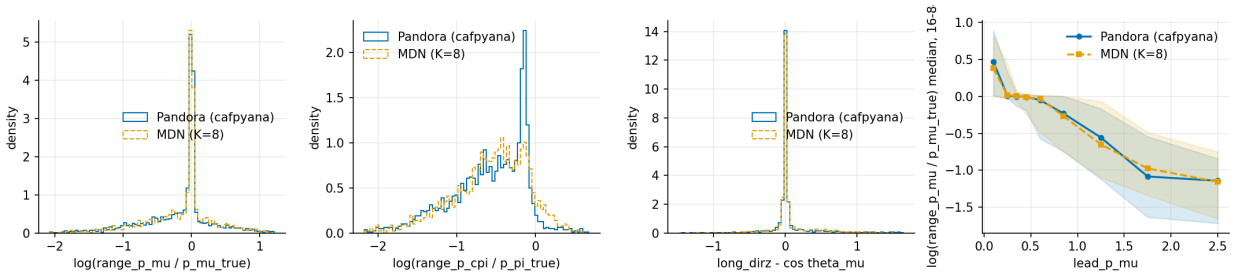


Figure 6: cohpi two-track response on 1mu1pi events passing the two-track topology.

5 M2: flow-matching surrogate of the slice block

Generation proceeds in three steps. (i) A *topology* head: a softmax classifier over the joint pattern of PFP counts (`n_pfp, n_prong, n_trk, n_proton, n_shower`; the 120 most frequent tuples plus "other", whose members are drawn from the training pool), sampled first; its one-hot encoding conditions everything that follows. (ii) Gradient-boosted heads for the presence of the variable groups that may not exist in a slice (second track, two-track block, hits within 1–4 cm of the vertex), for the two containment flags, and for the point masses at detector boundaries (vertex pinned at $z = 0$ or at the x, y walls, back-to-back track pairs). (iii) A conditional flow-matching model (linear path, Gaussian source, 6-layer velocity MLP of width 1024) over the standardised continuous block, masked per event to the groups and dimensions that exist (sim2reco `dim_mask`). Positive quantities enter as logarithms, direction cosines and the opening angle through artanh (and direction vectors are renormalised), standardisation is robust (median, MAD). Inverse transforms produce a dataframe with the production column names. Trained on 2,614,990 matched slices for 60 epochs (26 min on one RTX 3090).

Earlier versions and what they taught: v1 (unbounded cosines, counts as dequantised flow dimensions) closed the cohpi cut flow only to 45% in the signal region; v2 (bounded cosines, point-mass heads) to 74%, losing most at the exact-count stage; v3 (topology head) closes every stage within statistics (Table 4).

Group / flag	rate [%]	log loss (marginal)	log loss (GBDT)	AUC
short	61.0	0.6688	0.0000	1.000
two	8.1	0.2804	0.0018	1.000
s1	86.7	0.3915	0.3290	0.778
s2	95.3	0.1889	0.1471	0.835
s3	97.3	0.1243	0.0915	0.864
s4	98.2	0.0913	0.0640	0.884
is_muon_contained	29.5	0.6064	0.3039	0.929
is_cpi_contained	56.2	0.6855	0.2827	0.950
spike_vtx_z	12.9	0.3848	0.0920	0.986
spike_vtx_x	5.6	0.2152	0.0494	0.990
spike_vtx_y	5.4	0.2108	0.0612	0.984
spike_opening_angle	3.0	0.1336	0.0980	0.872

Table 3: Group presence and containment heads, test files.

Closure on the test files. Table **??** compares every generated column with Pandora (fraction present and Wasserstein-1 in transformed units; median W_1 0.016, maximum 4.908). Table 4 recomputes the cohpi cut flow on the generated and the real test slices, for all slices and for true signal; this is the kind of comparison the analysers will make on the full surrogated sample.

Column	present, real [%]	present, generated [%]	W_1 (transformed units)
n_pfp	100.0	100.0	0.0047
n_prong	100.0	100.0	0.0025
n_trk	100.0	100.0	0.0017
n_proton	100.0	100.0	0.0022
n_shower	100.0	100.0	0.0022

Column	present, real [%]	present, generated [%]	W_1 (transformed units)
vtx_x	100.0	100.0	0.4595
vtx_y	100.0	100.0	0.5235
vtx_z	100.0	100.0	0.4152
nu_score	100.0	100.0	0.0007
bcfm_score	99.2	100.0	0.0037
total_pe	99.6	100.0	0.0043
long_dirx	100.0	100.0	0.0038
long_diry	100.0	100.0	0.0066
long_dirz	100.0	100.0	0.0061
long_trk_len	100.0	100.0	0.0049
long_trk_pid	100.0	100.0	0.0010
long_trk_chi2_mu	100.0	100.0	0.0117
long_trk_chi2_pro	100.0	100.0	0.0165
long_trk_chi2_mu_new	100.0	100.0	0.0110
long_trk_chi2_pro_new	100.0	100.0	0.0169
short_dirx	61.0	60.9	0.0079
short_diry	61.0	60.9	0.0118
short_dirz	61.0	60.9	0.0103
short_trk_len	61.0	60.9	0.0142
short_trk_pid	61.0	60.9	0.0038
short_trk_chi2_mu	61.0	60.9	0.0173
short_trk_chi2_pro	61.0	60.9	0.0314
short_trk_chi2_mu_new	61.0	60.9	0.0165
short_trk_chi2_pro_new	61.0	60.9	0.0288
reco_t	8.1	8.2	0.0188
opening_angle	8.1	8.2	0.0688
range_p_mu	8.1	8.2	0.0056
range_p_cpi	8.1	8.2	0.0111
muon_start_x	8.1	8.2	0.9986
muon_start_y	8.1	8.2	1.3456
muon_start_z	8.1	8.2	3.2313
muon_end_x	8.1	8.2	2.6932
muon_end_y	8.1	8.2	2.7195
muon_end_z	8.1	8.2	4.9076
cpi_start_x	8.1	8.2	0.9846
cpi_start_y	8.1	8.2	1.3408
cpi_start_z	8.1	8.2	3.3058
cpi_end_x	8.1	8.2	1.5248
cpi_end_y	8.1	8.2	1.5108
cpi_end_z	8.1	8.2	3.8108
sum_integ_1cm	86.7	86.8	0.0108
sum_integ_2cm	95.3	95.5	0.0188
sum_integ_3cm	97.3	97.3	0.0143
sum_integ_4cm	98.2	98.1	0.0187

Generation. `scripts/generate_slc.py` loads the trained surrogate and writes slice records for any truth table: slice cardinality (0, 1 or 2 per neutrino) from its head, then topology, presence, point masses and the flow, then the production column names, plus the recomputed cohpi flags. On the 814,028 test neutrinos of the five chunks it generates 269,841 slices in 66 s on one GPU (about 12,000 neutrinos per second, so the full official sample takes a quarter of an hour), with the slice-multiplicity fractions (0 / 1 / 2 slices: 68.0 / 31.0 / 1.1%) reproduced to 0.1% and 422 slices

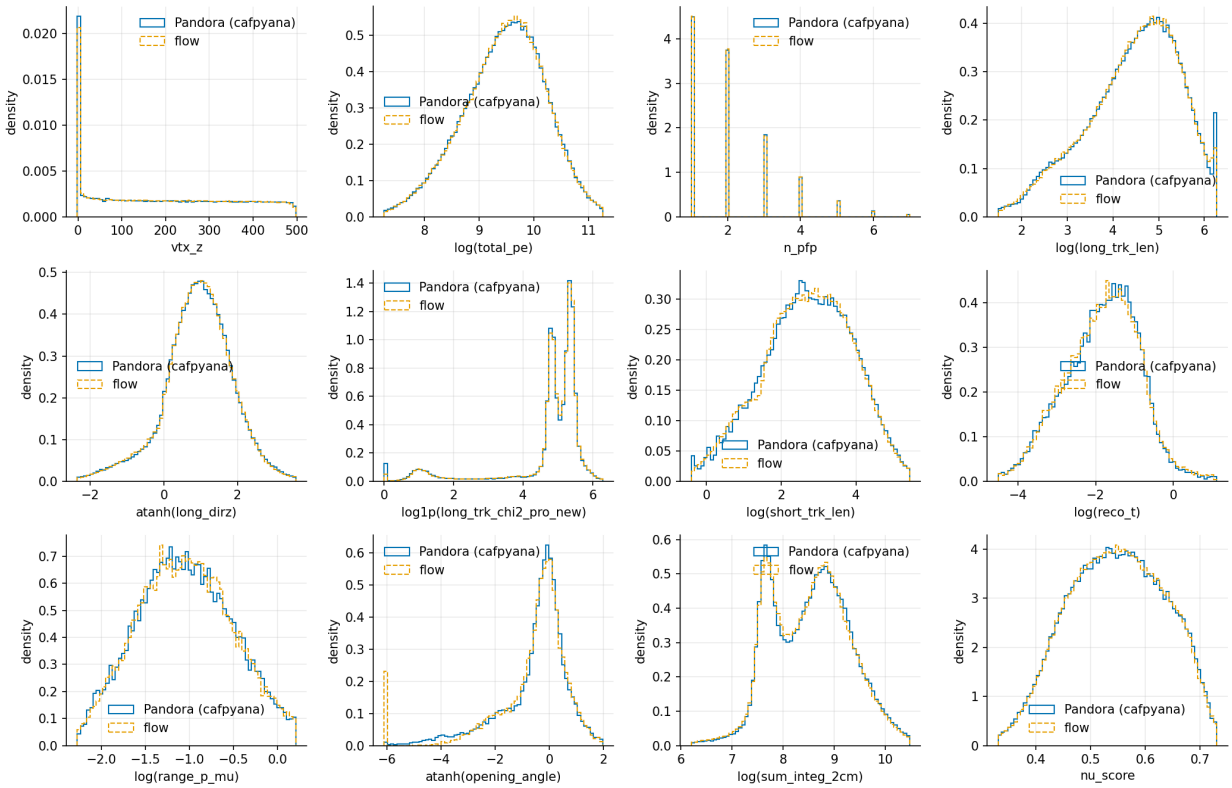


Figure 7: Marginals of generated (flow) against Pandora slice variables, test files.

passing the full cohpi selection against 423 for Pandora (216 against 220 in the signal region).

Stage	all, Pandora	all, generated	signal, Pandora	signal, generated
pe_bcfm	205,571	205,634	190	190
2trk	17,334	17,521	88	98
dir	1,467	1,560	75	84
chi2	862	952	65	77
sphere	504	529	62	69
cathode	423	434	58	62
sr	220	212	55	56

Table 4: cohpi cut flow recomputed on real and generated test slices.

6 Outlook

1. Official sample (9.94 M triggers): retrain M1/M2, add the `mcdaught` and full-PFP tables.
2. M3: particle-set encoder (transformer over primaries + vertex token) replacing the fixed features.
3. M4: PFP set generation (cardinality + equivariant set flow), Bayesian last layers for an epistemic flag.
4. Holdouts: train on $|t| < 0.1$, test the tail; low versus high E_ν; forward versus wide-angle muons; ≤ 2 versus ≥ 3 charged primaries; one TPC versus the other.
5. Surrogated dataframes written in the production format for the cohpi and kaon analysers.